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30 March 2026
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Dear Prof. Akira S Mori,

We would like to begin by thanking both you and the two anonymous reviewers for your creative and valuable thoughts. On behalf of myself and my coauthors, we appreciate your time and contributions.

Reviewer 1 was fully satisfied with our previous responses and complimented the new analyses included. Reviewer 2 was also highly supportive of our updated manuscript, and provided a clever solution for how we could resolve their major outstanding critique. Here, we have incorporated their advice, principally fitting generalized linear models to link the ideas of habitat usage and morphology more clearly. We have also incorporated your suggestions regarding adjustments to figures that we think improve their value.

Below please find **reviewer comments in green**, and our responses in black. New text added to the manuscript is **highlighted**. We have formatted a page break between each reviewer. Where necessary, we have interjected our responses among reviewer comments.

On behalf of myself and coauthors, we thank you for your time and consideration,

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Referee 1:

I have reviewed the response to reviewers, and have read the revised manuscript. I believe the authors did an excellent job in responding to reviewer comments and concerns. I thought the additional analyses of niche breadth evolution made sense and improve the manuscript in providing needed quantitative analysis to support their hypotheses, which they now more carefully lay out. I think this article will be of interest to a broad readership in ecology and evolution.

We appreciate the time and energy you have taken to review our manuscript a second time. Thank you for your enthusiastic support of our revised manuscript and the additional analyses included. We hope readers of Ecology Letters enjoy it similarly.

Referee 2:

The revised manuscript improves upon the previous draft, particularly by quantifying generalization and presenting analyses regarding how generalization proceeds through the phylogeny. I think this is a valuable contribution, and nicely illustrates important concepts at the interface of ecology and evolutionary biology. However, reading throughout the manuscript I still find myself wanting the final pieces of the argument that the authors are making to be rigorously tested from the data.

Thank you for taking the time to thoroughly review our manuscript for a second time. We appreciate the energy you have invested in improving this work. We also believe this is a valuable contribution regarding paths towards specialization and believe it will be of interest to the ecology and evolutionary biology communities.

The most essential piece that is missing is a direct link between the idea of location in morphospace and the idea of ecological generalization. Currently there is only a verbal argument that species in the center of morphospace are generalists, but this idea is not actually tested.

We appreciate this point and agree that the link between morphospace position and generalization (via substrate usage breadth) must be better connected. This appears to be the primary consideration in this review (mentioned elsewhere below) and so we address it extensively here.

Conceptually, it is hard to define what the true “center” of the morphospace is. Is it the geometric middle (mean, median)? In which case the center will be skewed by imbalances in species richness. In our case the mean and median are pulled strongly towards the “terrestrial” morphotype due to large numbers of *Tympanocryptis* and *Ctenophorus*. Based on our modelling and the nested nature of these taxa, they are likely highly derived however. Instead, we have taken the midpoint (average of minimum and maximum values) of each morphological trait and used those to determine the morphological center.

Using this morphological midpoint we start by estimating the multivariate morphological distance between each tip/node and the midpoint, as the Mahalanobis distance. We follow the advice you provided in another comment below to fit generalized linear models, and carried this out using two approaches.

(1) We applied morphological distance as a predictor for niche breadth (the sum of substrates used) in a generalized linear model both with and without phylogenetic correction incorporated via the variance-covariance matrix of both living and ancestral taxa (internal nodes). Models were fit using the *stats* and *glmmTMB* packages in R. We followed the methodology of Williams et al. (2026) in applying the *glmmTMB* model.

```
glm(breadth ~ mahalanobis, data=data, family="poisson")
glmmTMB(breadth ~ mahalanobis + (1|species) + propto(0 + species|g, agam.vcv),
        data = bdist, family = "poisson", REML = F)
```

In the *glmmTMB* model the variance-covariance matrix is represented by the 'agam.vcv' object in the propto structure. Both phylogenetic and non-phylogenetic models indicated a significant effect of morphological distance on habitat breadth ($p=0.000205$; $p=0.000158$) with a negative relationship (-2.9155 ; -2.44116). We also fit an intercept-only model ($\text{glm}(\text{breadth} \sim 1 \dots)$; $\text{glmmTMB}(\text{breadth} \sim 1 \dots)$) to determine relative fit of the data.

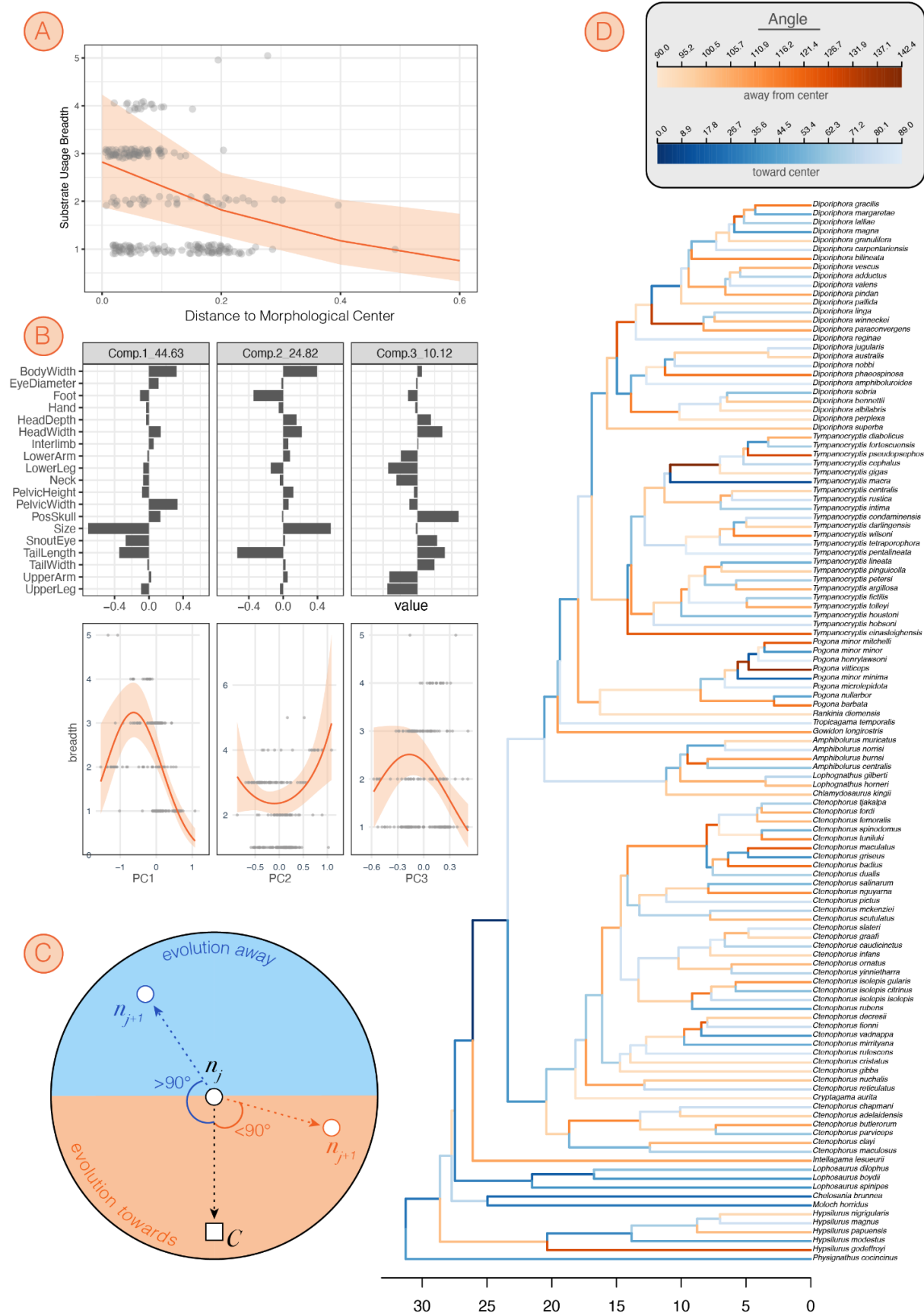
(2) As a comparison, we also fit phylogenetic and non-phylogenetic GLMs to the first three principal component axes, which combined accounted for ~80% of the variation. Following your recommendations, to the predictor variables we applied a quadratic function to explain the expected hump-shaped response of habitat breadth (lower at high and low PC scores, highest at intermediate scores).

```
glm(breadth ~ PC1 + I(PC1^2) + PC2 + I(PC2^2) + PC3 + I(PC3^2), family = "poisson",
data = data)
glmmTMB(breadth ~ PC1 + I(PC1^2) + PC2 + I(PC2^2) + PC3 + I(PC3^2) + (1|species)
+ propto(0 + species|g, agam.vcv), data = data, family = "poisson",
REML = F)
```

Both phylogenetic and non-phylogenetic models indicated significant effects of PCs on habitat breadth (see Supplement *GLM Model Fitting*). These are visualized in Figure S15 and that figure is reproduced on the next page. Interestingly, the expected hump-shape responses are visible in PCs 1 and 3, but PC 2 shows a non-significant ($p=0.43$) inverse relationship (U-shaped). PCs 1 and 3 show heavy loadings for traits typically associated with niche divergence, such as size, tail length, body width, and limb lengths. The turning points (quadratic vertices) of PCs 1 and 3 are close to the estimated empirical midpoints of those components (PC1: -0.63 vs -0.22 ; PC3: -0.16 vs -0.04), indicating that breadth decreases away from the morphological midpoints.

Lastly to help visualize evolutionary trajectories relative to the morphological center, we created a branch-by-branch summary of morphological trends. To do this we measured the angle of morphological change between each node, their direct ancestral node, and the morphological center/midpoint (see Fig.S15C). We classified trajectories with angles $<90^\circ$ as moving *towards* the morphological center, and classified trajectories with angles $\geq 90^\circ$ as moving *away* from the morphological center. For example, an acute angle of 5° is evidence of a strong move towards the hub of the morphospace, while an obtuse angle of 175° is evidence of a strong move away from the hub. The color and saturation of each branch indicates the direction and value of the angle of change (Fig.S15D). Early branches of the tree predominantly show movement towards the center of morphospace.

We thank you for the creative GLM method suggested and believe these approaches and results now better link morphospace position and niche breadth.



Supplementary Figure accompanying GLMMs. (A) Plot of habitat usage breadth as a function of Mahalanobis distance to the morphological center. Breadth scores are integers and jitter around values is solely to improve visualization of concentration of points. Orange line indicates best fit prediction

resulting from the phylogenetic generalized linear mixed model. (B) Three panels indicate trait loadings on the first three Principal Component axes that account for ~80% of trait variation. (C) Corresponding plots of habitat usage breadth against PC scores, with best fit GLM lines and confidence intervals overlaid in orange. (D) Demonstration of how the angle of morphological change was calculated between an ancestral node (n_j), descendant node (n_{j+1}), and the morphological center/midpoint (C). Angles $<90^\circ$ were classified as movement *towards* the center, angles $\geq 90^\circ$ were classified as movement *away* from the center. Color and saturation of each branch indicates the direction of movement and value of the angle.

L81: “with generalist morphologies”. Still unclear what is meant by a generalist morphology or why being at the center of morphospace would be expected to confer generalism. Could be spelled out by the assumption that specialists need extreme morphologies, and something in the middle presumably has some access to all the ecologies that these extremes grant. Other potential rationales probably exist. The logic behind the center of morphospace corresponding to generalist species needs to be made clear here.

We agree that this could be further clarified. To address this comment we have amended the text to more explicitly state the hypothesis following your suggestion:

The expectation being that ecological specialists require extreme morphologies that evolve from ancestors with broad niche breadths and morphologies nearer to the morphospace center.

L106: “To address the evolution of the amphibolurine adaptive phenotypic landscape,” For the readership of a venue like Ecology Letters this somewhat vague statement should be replaced by specific hypotheses that are being tested.

We have changed this sentence to be more in line with our stated hypothesis:

To address the idea that specialist peaks in the adaptive phenotypic landscape are linked through generalist ancestors, we first reconstruct the phylogenetic relationships of Australo-Papuan dragon lizards.

L189: Describe what “low structure”, “high structure” mean in the main text, or give examples. I would have thought high structure was the same as arboreal based on word choice. Based on the supplement, something like “low structure (e.g. vegetation shorter than $<30\text{cm}$), high structure (vegetation $>30\text{cm}$ and less than X), arboreal (vegetation $>X$)” would work. Some argument needs to be given that these categories represent something meaningful about how lizards use the environment, such that generalization can be meaningfully measured by counting categories. Basically you need to be able to convince a light skeptic that a species that uses arboreal and high structure vegetation (i.e. 2 categories) is more specialized than one that uses rock, terrestrial, and low structure (i.e. 3 categories), and that this is not just an artifact of an arbitrary decision about how to divide up the categories.

Unfortunately, Ecology Letters applies a strict word limit to the main text. We appreciate the request for more information, but believe the current level of detail is appropriate for the main text and most readers. We refer readers who have further interest to the supplement for more details.

291: This analysis is an absolutely great addition. Some indication of statistical support for these models over alternatives is needed to actually test the relevant hypothesis though.

Thank you for catching this, it is important to provide context for the level of support. AIC scores and AIC weights are included in Supplementary Figure S13. In the text we have indicated the AIC weights.

Fig 2: Unclear what the numbers above the arrows are: Are these rates or support values (e.g. p-values) associated with the arrows. Explain what these numbers mean in the legend. Make it more clear which node corresponds to the ancestor of the Australian clade as described in the figure legend. It's implied that it's the large one labeled 3+ in the lower panel, but add a label there to clarify the figure. It's not clear what the "1" is trying to communicate pointing at the orange/tree section of the pie chart.

The numbers above the arrows indicate the evolutionary transition rate between states. We have added a sentence to the legend to indicate this.

More discussion of the alternative models tested is needed, so the reader can evaluate that the relevant model space is being considered. It's unclear whether you evaluated unequal transitions from the 3+ state and into the 3+ state, when it looks like (and is implied by the previous model) that 3+ should go to singletons much more than singletons should go to 3+. Also it looks like ground transitions into rock a number of times (a plausible transition ecologically), but in the figure there's no indication that this transition is more likely than other singleton transitions. More transparency about what was done is needed here.

We understand the request for more information about alternative models, but are limited in our ability to expand on these in the main text. In the Supplement, there is a section titled *Niche Breadth Models* (pg. 6-7) that outlines the models compared. Paired with Fig.S13, we believe there is sufficient information for readers to deem our model selection and comparison appropriate. We refer readers to the Supplement in the main text ("see Supplement for more details") but are unfortunately constrained by a strict word limit.

L346: "This exploration was driven first by a transition from an arboreal to a generalist ancestor". As the editor pointed out, the support for an arboreal ancestor is not overwhelming: 66% of being in the arboreal regime is only slightly better than a coin flip. Or even in the new models where it's around 75% for the overall ancestor being 'tree' associated and the Australian 1 being 3+/generalist, the specific transition from Arboreal to generalist will only happen less than 2/3rds of cases assuming independence ($0.75 * 0.75 = \sim 0.53$). I think the issue here can be solved by weakening the language slightly.

This is a fair critique. To accommodate it, we have added the phrase "we argue" to tone down the language slightly and emphasize this is our interpretation. The sentence now reads

■ We argue this exploration was driven first by a transition from ...

L382: Part 1: "intermediate forms that use a variety of substrate types" This analysis is missing from the paper. I would like to see evidence that morphologically intermediate forms are the

ones that use a variety of substrate types. Specifically this analysis would take the form of something like a quadratic poisson glm (in R pseudo code):

```
Glm(substrate.number.used ~ morph.pc1 + l(morph.pc1^2) + morph.pc2 + l(morph.pc2^2) +  
[optional: morph.pc1:morph.pc2], family='poisson')
```

This is a great suggestion and a methodology that accurately links the separate ideas of morphological centrism and niche breadth. We address this idea extensively above (see response to second comment from R2). For quick reference, however, we have added the following text to the Methods section:

Linking Morphology to Niche Breadth

To test the idea that niche breadth is dictated by morphological position, we used phylogenetically corrected Generalized Linear Mixed Models in *glmmTMB* (Brooks et al. 2017) following the practice of Williams et al. (2026). We fit two model types with differing predictors (see Supplement for details) using the sum of substrate as the response. In the first, we estimated the multivariate distance to the morphological midpoint as the Mahalanobis distance, and in the second we fit the first three Principal Component axes (accounting for 80% of variation) including quadratic terms to account for expected hump-shaped responses as PC scores become more extreme.

We have added the following to the Results section:

Morphology and Niche Breadth

Phylogenetic GLMMs indicate significant effects of distance to the morphological center and PCs 1 and 3 on predicting substrate usage breadth. Niche breadth declines with increasing distance from the morphological center (-2.19, $P < 0.01$). PCs 1 (-0.79, $P < 0.001$) and 3 (-2.19, $P < 0.01$) show negative quadratic functions where substrate breadth is highest at intermediate PC scores.

With significant quadratic terms, specifically with negative coefficients (plus plotting the best fit line showing that predicted substrate useage is indeed highest for intermediate morphologies), taken as evidence supporting the hypothesis/statement.

Connecting the morphology with the ecological data is the main missing link in this manuscript as it stands. [You would potentially also want to take phylogenetic non-independence into account, using the various pglmm capable packages].

Particularly important from the point of your argument in the discussion is showing that the predicted morphologies of the species inferred to be present in the past (i.e. the immediate ancestors in figure 5) are predicted to use more substrates (based on their morphologies) than their decedents.

This is another important idea. Not just that we see this relationship in living taxa, but are able to identify the process in ancestral taxa too. To make sure this is incorporated appropriately—and in a phylogenetic context—we generated a variance covariance matrix for the amphibolurine tree including ancestors using *phytools* function *vcvPhylo* with the argument *anc.nodes=T*. This allowed us to include the estimated morphologies, morphological distances, and niche breadths of ancestral taxa.

Doing this actually ties together your argument. Otherwise what you are showing is that lizards have evolved from generalist ancestors into specialists decedents (on average), and also, but potentially unrelatedly, lizards have typically evolved out from the center of morphospace (after initially moving towards the center in the colonization of Australia). That the morphological regimes basically entirely match the substrate useage data as mapped on the phylogeny strongly implies that there is a link, and then if you also look at figure 5 you can kind of see the central morphology bit. But this requires the reader to look across 3 different figures (which when in printed/pdf form will be on different pages) and correlate the contents for themselves. There needs to be a single analysis that actually supports the relationship between these 3 ideas (morphology, generalization, and trajectory of evolution) with statistical evidence such that the idea is falsifiable.

We very much agree. We have addressed the additional analyses (GLMMs) above in our response. Addressing the visualization of these patterns is trickier. To show how morphology predicts niche breadth we have adapted Figure 5 to include plots from our GLMMs. These show the trends in niche breadth as a function of PC1 and distance to the morphological center. We have also used the fitted coefficients to predict breadth across a biplot of PCs 1 and 3 to show the breadth-surface. We have incorporated these changes into the figure legend:

Niche breadth decreases as morphologies become more extreme through changes in suites of traits. Plots in top row show results of GLMMs. (Left) niche breadth is greatest at intermediate PC 1 scores. (Center) niche breadth decreases with increasing distance from the morphological center. (Right) visualization of predicted breadth values (background colors) plotted against PCs 1 and 3.

L382: Part 2: “However, intermediate forms that use a variety of substrate types (Amphibolurus, Gowidon, Lophognathus, Tropicagama, et al.) fill the morphological space, uniting disparate peaks of the landscape and facilitating transitions to new ecologies.” Someway to communicate this to readers who are not lizard taxonomists is needed here. (i.e. referencing a panel of a figure showing the placement of these genera in morphospace).

This is a valuable point. We have included a reference to Figure 1B, and have also added small animal silhouettes to Figure 5 to guide readers visually.

L434: “This provides more evidence that transitions across the landscape from specialist to specialist are uncommon and likely require an intermediate form (Fig.2) (Dennis et al. 2011).” I think it’s fair to say given the data that it’s relatively unlikely that a specialist will backtrack to become a generalist if generalists are already present in the fauna, and so transitions are most likely to be into specialists from the pre-existing generalist clade. I think it’s worth mentioning this because it provides some additional insight on how evolution proceeds when niche incumbency is an issue. If I can paraphrase what I’m taking away from your data, you will fill niches by moving from generalists -> specialists. You are unlikely to go from specialist -> generalist, and therefore are unlikely to go from specialist -> generalist -> specialist, except in circumstances of overwhelming ecological opportunity (e.g. colonization of Australia). Therefore the generalists in the center of morphospace (assuming there ends up being statistical support for this proposition) act like an anchor. There’s this anchor of generalists

ready to spit out additional specialists if the opportunity arises, whereas the specialists mostly stay in their lane (minus maybe the transition from ground to rock... but rock is kind of just a special type of ground?). A suggested possible way to integrate these ideas: “from specialist to specialist are uncommon and likely require an intermediate form. However the existence of large generalists clades throughout evolutionary history of this group precludes frequent backtracking to intermediate morphologies by specialists. Instead, pre-existing generalists with intermediate morphologies will pump out additional specialists.”

We agree in principle that the highly specialized taxa (breadth score of 1) are unlikely to evolve towards absolute generalists (breadth score of 5). In fact, the best fitting model (Fig.2, Fig.S13K) explicitly does not allow these types of transitions ($1 \leftrightarrow 5$) because we considered these to be biologically unrealistic. Instead, the two best fitting models (Fig.S13J/K; NJ-ER/NJ) allow a loose stepwise process in which only one or two states can be gained or lost at a time. This is preferred over strict stepwise models (Fig.S13H/I; STP/STP-ER). The model you verbally describe is most similar to a specialization process in which transitions back towards increasing breadth are limited. We fit a model of this description as well, Fig.S13M (SPC), however, there is limited support for it (AICw 10%).

We disagree that rock is “just a special type of ground”, as the category is meant to capture species with saxicoline habits. This habitat often (but not always) selects for a unique morphotype that is dorsoventrally flattened with long limbs and large feet/hands (see *Ctenophorus ornatus* and *C. rufescens*). These are traits captured in our morphological measurements and reflected in our analyses of niche breadth. Regardless, we appreciate your thoughts here.

Fig S8: No explanation of what red lines are.

This figure is effectively a reproduction of Fig.1C, with the species richness trajectory included. The lines represent simulated trajectories under a null model. We have adjusted the figure legend to now read:

The Amphibolurinae hypervolume expands early and late in their evolution, with a period of niche packing in the Miocene. Diversification and morphological expansion are coupled in the early radiation of dragon lizards, but are largely decoupled from the Miocene onwards. Green line represents the accumulation curve for species, black line the hypervolume through time, and red lines are 500 null expectations.

In supplement about habitat classification: “due to different physical requirements resulting from sheer forces.”. Unclear what you mean here. Describe “sheer” forces in more depth and the ecology behind what you’re describing.

Prior studies within dragons have emphasised the importance of habitat usage on morphology, we refer to those here (Stuart-Fox & Owens 2004; Collar et al. 2010). The principal consideration is that different surfaces have different requirements. We expand on this in the supplement with additional lines that read:

For example, narrow branches and twigs in low and high structure allow for ample grasping and are often oriented horizontally. In contrast, trees and rock outcrops provide broad or vertical surfaces that may require different morphologies (longer limbs, claws) to access them. Evidence for this partitioning among substrate types has

been documented for Australian dragons (Stuart-Fox & Owens, 2004; Collar et al. 2010).

Some what weird that “rock” is both its own habitat category, and a sub category within the “low structure” category. I think I understand the difference that you’re likely thinking of, but you should describe it more clearly in the supplement.

The supplement describes that the “Rock” category encompasses “exposed rock surfaces and outcrops with little-to-no vegetation”, which is fundamentally a different surface to a stone or rock that is embedded in another landscape (as mentioned in the “Low Structure” category). To specify this, we have changed the wording in the “Low Structure” category to read “stones” instead of “rocks”, and the wording in the “Ground” category to read “stony plains”, so as to not confuse readers.

Editor:

This revised version has incorporated many of the comments and concerns from the two initial reviews, and both reviewers remain excited by the study's aims, data, and overall analytical framework. For instance, the reconstruction of ancestral ecologies using reconstructed morphologies via RF does seem novel, and potentially informative. However, reviewer two *still* finds the manuscript more descriptive than hypothesis-driven. This is not a standard Ecology Letters study, but, given it has been submitted here, then, for Ecology Letters, we need clear hypotheses linked to clear predictions, clear tests, and clear, statistically-supported support/refutation. I agree with the overall assessment and the specific recommendations.

Thanks for the opportunity to resubmit our work. We think the revised version has a greater emphasis on testing the ideas, including a better methodological backing via the GLM exercise (thank you R2). We are excited by the idea of presenting something that is a slightly different flavor, to the Ecology Letters community.

Importantly, I see nothing in the reviewer's suggestions that cannot be dealt with. The sensitivity of the reconstructions to the scoring of habitat use and the mapping of morphology to "generalism" are both straightforward. Clarification of some of the results (e.g. the numbers on the transition arrows) and some statistical support for claims can be done easily.

We have responded to all reviewer comments and incorporated all requests as best as possible. Where those could not be accommodated in the main text, we have further fleshed out details in the Supplement.

A few additional minor things from me wrt to figures:

Figure 1: Could you point out the Australian MRCA here? And perhaps the support sentence at line 506 should be moved up?

We have added labels to Figure 1 to indicate what we are now calling the "Australian MRCA" and the "Core Australian" groups, as these otherwise prove confusing. We have also moved the comment about support values to the beginning of the figure legend.

Figure 2: Could the big pie charts be moved off of the nodes so we can see which they refer to (like in Fig. 3)? The numbers pointing to a few of these larger pie charts are also not explained in the caption.

We have moved the pie charts off the nodes and clarify additional details for this figure in the legend, including numbers near large pie charts and the inset models.

Figure 3: Is the reconstruction of the MRCA regime valid given there are no outgroup morphologies considered? Or were they and they are just not shown?

We consider the ancestral regime estimates valid despite the fact that we did not include outgroup representatives from the Draconinae, Agaminae, and Hydrosaurinae in our morphological model fitting. While phylogenetic models are influenced by surrounding states, the random forests approach we used should be insensitive. Instead, it predicts assignments based exclusively off the morphological data presented and is agnostic to phylogeny. Incorporating additional subfamilies is an exciting proposal and one that would provide insight into the processes and patterns of agamid evolution as a whole. It's an idea that we would love to follow up on in the future, but is beyond the scope of this current project.

Figure 4: Do you really mean "paradigm shift"? Clades have neither "approaches" nor "assumptions."

We are not entirely sure we understand the comment made here. We use the term “paradigm shift” to describe the magnitude of the change that takes place at this point along the amphibolurine evolutionary history. Alternatively we could use the phrase “turning point” or something similar, but we believe the phrase as written best encapsulates the dramatic morphological and ecological change that occurred at this point. We agree that clades have neither “approaches” or “assumptions” but are not sure how this connects with the previous comment.

I look forward to sending a new version to reviewer 2 for a quick once-over in short order.

Thank you for your interest and support of our work. We believe that the changes described above have strengthened the manuscript and appreciated the input of the reviewers and yourself.